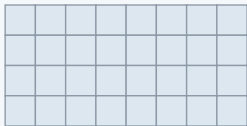


## Flattened Representation

4 × 8 physical UPA



4 rows × 8 columns



Flatten

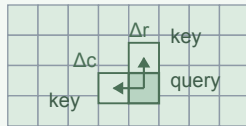
$$u = 0, 1, \dots, 31$$

$$u = rN_v + c$$

Row-column topology is not explicitly encoded in the model interface

## Explicit UPA Representation

4 × 8 row-column topology



$$(t, k, u) \text{ vs. } (t, k, r, c)$$

$r$  = row,  $c$  = column

Explicit row-column topology

UPA geometry is explicitly represented